Problem 1

Find the Maximum Likelihood Estimation (MLE) of θ in the following probabilistic density functions. In each case, consider a random sample of size n. Show your calculation:

1a

$f(x|\theta) = \frac{x}{\theta^2} exp\{\frac{-x^2}{2\theta^2}\}, x \geq 0$
$L(\theta|x) = ln(\prod_{i=1}^{n} \frac{x}{\theta^2} exp\{\frac{-x^2}{2\theta^2}\})$
$L(\theta|x) = \sum_{i=1}^{n} ln(\frac{x}{\theta^2} exp\{\frac{-x^2}{2\theta^2}\})$
$L(\theta|x) = \sum_{i=1}^{n} ln(x) - log(\theta^2) - \frac{x^2}{2\theta^2}$
$L(\theta|x) = \sum_{i=1}^{n} ln(x) - 2log(\theta) - \frac{x^2}{2\theta^2}$
$L(\theta|x) = \sum_{i=1}^{n} ln(x) - \sum_{i=1}^{n} 2ln(\theta) - \sum_{i=1}^{n} \frac{x^2}{2\theta^2}$
$L(\theta|x) = \sum_{i=1}^{n} ln(x) - \sum_{i=1}^{n} 2ln(\theta) - \frac{1}{2\theta^2} \sum_{i=1}^{n} x^2$
Take the partial derivative and set it equal to zero.
$\frac{\delta L(\theta|x)}{\delta\theta} = 0 - \frac{2N}{\theta} + \frac{1}{\theta^3} \sum_{i=1}^{n} x^2$
$0 = -\frac{2N}{\theta} + \frac{1}{\theta^3} \sum_{i=1}^{n} x^2$
$\frac{2N}{\theta} = \frac{1}{\theta^3} \sum_{i=1}^{n} x^2$
$2N\theta^2 = \sum_{i=1}^{n} x^2$
$\theta = \frac{1}{\sqrt{2N}} \sqrt{\sum_{i=1}^{n} x^2}$

1b

$f(\alpha, \beta, \theta|x) = \alpha\theta^{-\alpha\beta} x^\beta exp\{-(\frac{x}{\theta})^\beta\}, x \geq 0, \alpha > 0, \beta > 0, \theta > 0$
$L(\alpha, \beta, \theta|x) = \sum_{i=1}^{n} ln(\alpha\theta^{-\alpha\beta} x^\beta exp\{-(\frac{x}{\theta})^\beta\})$
$L(\alpha, \beta, \theta|x) = \sum_{i=1}^{n} ln(\alpha) + ln(\theta^{-\alpha\beta}) + ln(x^\beta) - (\frac{x}{\theta})^\beta$
$L(\alpha, \beta, \theta|x) = \sum_{i=1}^{n} ln(\alpha) - \alpha\beta ln(\theta) + \beta ln(x) - (\frac{x^\beta}{\theta^\beta})$
$L(\alpha, \beta, \theta|x) = \sum_{i=1}^{n} ln(\alpha) - \sum_{i=1}^{n} \alpha\beta ln(\theta) + \sum_{i=1}^{n} \beta ln(x) - \frac{1}{\theta^\beta} \sum_{i=1}^{n} x^\beta$
Take the partial derivative and set it equal to zero.
$\frac{\delta L(\alpha,\beta,\theta|x)}{\delta\theta} = 0 - \frac{N\alpha\beta}{\theta} + 0 + \frac{\beta}{\theta^{\beta-1}} \sum_{i=1}^{n} x^\beta$
$\frac{N\alpha\beta}{\theta} = \frac{\beta}{\theta^{\beta-1}} \sum_{i=1}^{n} x^\beta$
$\frac{\theta^{\beta-1}}{\theta} = \frac{1}{N\alpha} \sum_{i=1}^{n} x^\beta$
$\theta = (\frac{1}{N\alpha} \sum_{i=1}^{n} x^\beta)^{\frac{1}{\beta-2}}$

1c

$f(x|\theta) = \frac{1}{\theta}, 0 \leq x \leq \theta, \theta \geq 0$
As θ can equal x, when $\theta = x$ then $f(x|\theta) = 1$
$\theta = x$

Problem 2

Bayes' Rule is made up of a posterior, prior, likelihood, and evidence.

$$P(C|x) = \frac{P(C)p(x|C)}{p(x)}$$

2a

Assuming equal priors, and $x = 2.5$,
$P(x|C_1) = \frac{1}{6}$ and
$P(x|C_2) = \frac{1}{4} * (2.5 - 1) = 0.375$
Since C_2 has a higher discriminant function value, this instance should be classified as such.

2b

Assuming different priors we have the following two equations:
$P(C_1|x) = 0.7 * \frac{1}{6} = 0.117$ and
$P(C_2|x) = 0.3 * \frac{1}{4} * (5 - 4) = \frac{3}{40} = 0.075$
Since C_1 has a higher score, this instance should be classified as such.

2c

Consider the point at which the two density functions cross:
The density function for class 1 is uniform at $y = \frac{1}{6}$
Using $\frac{1}{6}$ as the y-value, $\frac{1}{6} = \frac{1}{4}(x - 1)$. We use $\frac{1}{4}(x - 1)$ because symmetry exists over $x = 3$ and we can simply use that symmetry later.
and we find $(\frac{5}{3}, \frac{1}{6}$ as the point where the functions cross.
Now, consider that density function area between $x = 1$ and $x = \frac{5}{3}$. Half of that area could be miss-classified as class 1 when actually class 2 and vice versa. To minimize the miss-classification that could occur here, we set α to be the mid-point between $x = 1$ and $x = \frac{5}{3}$, and thus

$$\alpha = \frac{4}{3}$$

Now, consider that, by definition of the problem, values between $3 - \alpha < x < 3 + \alpha$ are classified as class 2. However, there exists a triangular area within the density functions that will be miss-classified with this definition and $\alpha = \frac{4}{3}$

These triangles have base length of $\frac{1}{3}$ and height of $\frac{1}{12}$ and two exist on either side of $x = 3$. The four triangles together have total area $\frac{1}{18}$. The total area under the density functions are $\frac{1}{6} * 6 + \frac{8}{3} * \frac{1}{3} * \frac{1}{2} = \frac{22}{18}$

As a proportion of the total area under the density functions, miss-classification makes up $\frac{1}{22}$. Thus there is a $\frac{1}{22}$ or

$$4.5\%$$

chance of miss-classification.

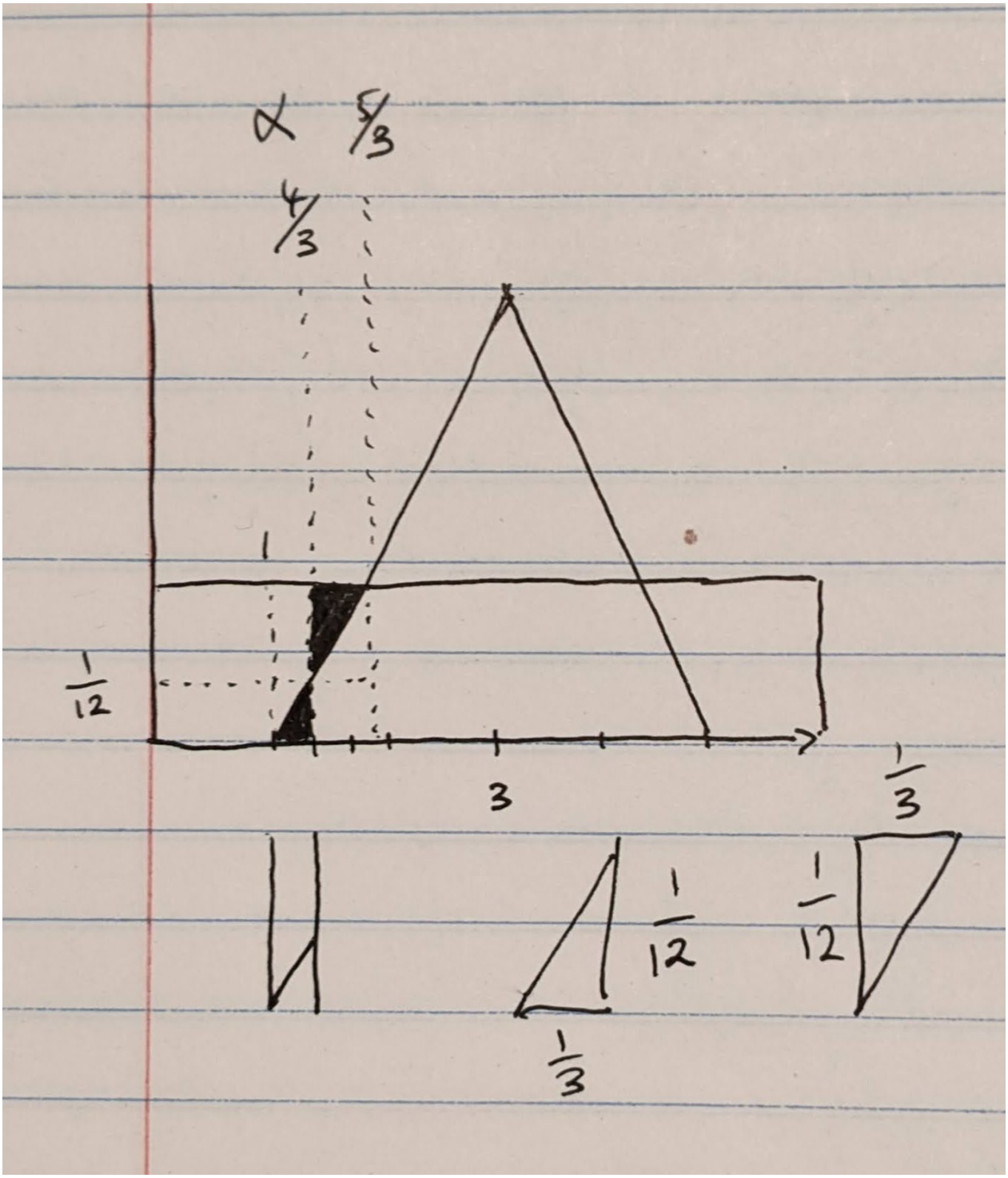


Figure 1: A graphical example of what areas are examined in 2C

Problem 3

Question 1

What is the discriminant function in each case? Show in your report and briefly explain.

Assume S1 and S2 are learned independently (learned from the data from each class):

$g_i(x) = -\frac{1}{2}log(|S_k|) - \frac{1}{2}(x_i - m_k)^T S^{-1}(x_i - m_k) + log(\hat{P}(C_i))$

Since the co-variance matrices are learned differently, they are important to the discriminant function and are present in the first term. Class means are always class specific.

Assume S1 = S2 (learned from the data from both classes):

$g_i(x) = -\frac{1}{2}(x_i - m_k)^T S^{-1}(x_i - m_k) + log(\hat{P}(C_i))$

Since co-variance matrices are calculated from the dataset as a whole without regard for labels, the classes share a co-variance matrix and the first term used, in the independently learned co-variances case are not needed.

Assume S1 = S2 (learned from the data from both classes), and the co-variance is a diagonal matrix:

$g_i(x) = -\frac{1}{2}\sum_{j=1}^{d}(\frac{x_j^t - m_{ij}}{s_j})^2 + log\hat{P}(C_i)$

The diagonal co-variance matrix assumes that features are not correlated to each other. The diagonal matrix can then just be calculated as a variance vector representing the variance across each feature.

Question 2

The confusion matrices show how many were correctly predicted and how many of each class label were predicted as the other class.

Report the confusion matrix on the test set for each assumption. Briefly explain the results.

Confusion Matrix for Gaussian Discriminant with class-dependent co-variance: $\begin{bmatrix} 14 & 5 \\ 16 & 65 \end{bmatrix}$

Confusion Matrix for Gaussian Discriminant with class-independent co-variance: $\begin{bmatrix} 29 & 9 \\ 1 & 61 \end{bmatrix}$

Confusion Matrix for Gaussian Discriminant with diagonal co-variance: $\begin{bmatrix} 26 & 11 \\ 4 & 59 \end{bmatrix}$

This shows that Gaussian discrimination with class-independent co-variance is best. Our assumptions for diagonal co-variance, where features are independent of one another, is not accurate as the scores for it are not as good as class-independent co-variance. This must mean that the features are somewhat correlated. The class-dependent co-variance assumption also did not perform as well as class-independent most likely because there were two distinct classes and they had different co-variance, so learning that co-variance and using it in the

discriminant yielded more accurate results than calculating a dataset-wide co-variance.